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ECPE 136
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Lab Assignment 8 - 2:1 Mux & 2ip Nand

Objective:

- The lab assignment's objective is to design a two input nand gate layout and a 2:1 mux layout with cadence virtuoso. We need to do DRC and LVS checks to verify that our design is correct and matches our schematic we designed earlier. Once completed we are to create a waveform from the Assura Quarus and analyze to make sure it is correctly representing the inputs and outputs.

Procedures:

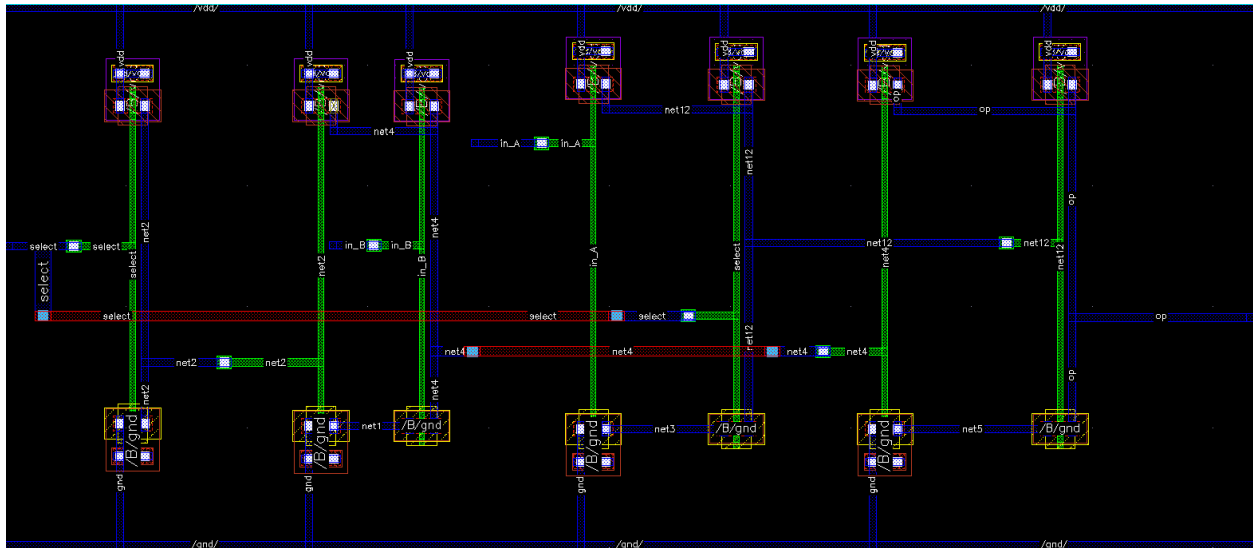
1.0 - Create a Layout

- Using the previous labs we will select the schematic 2-input NAND. From there we will open Layout XL to create a layout window. We can generate a layout with all variables and transistors from which we can then start making connections. With this layout we need to make the nmos gates in series and the pmos gates in parallel. Once all is completed we need to run a DRC and LVS check to verify the layout matches the schematic so a waveform can be generated.
- Again using the previous lab we will make the layout from the 2:1 mux schematic previously created. With this we need to make sure that we create an inverter and three two input NAND gate layouts. With this each NAND will have one variable of a or b followed by a select or select'. Those two nands will output to another NAND which then does a final output to the main output.

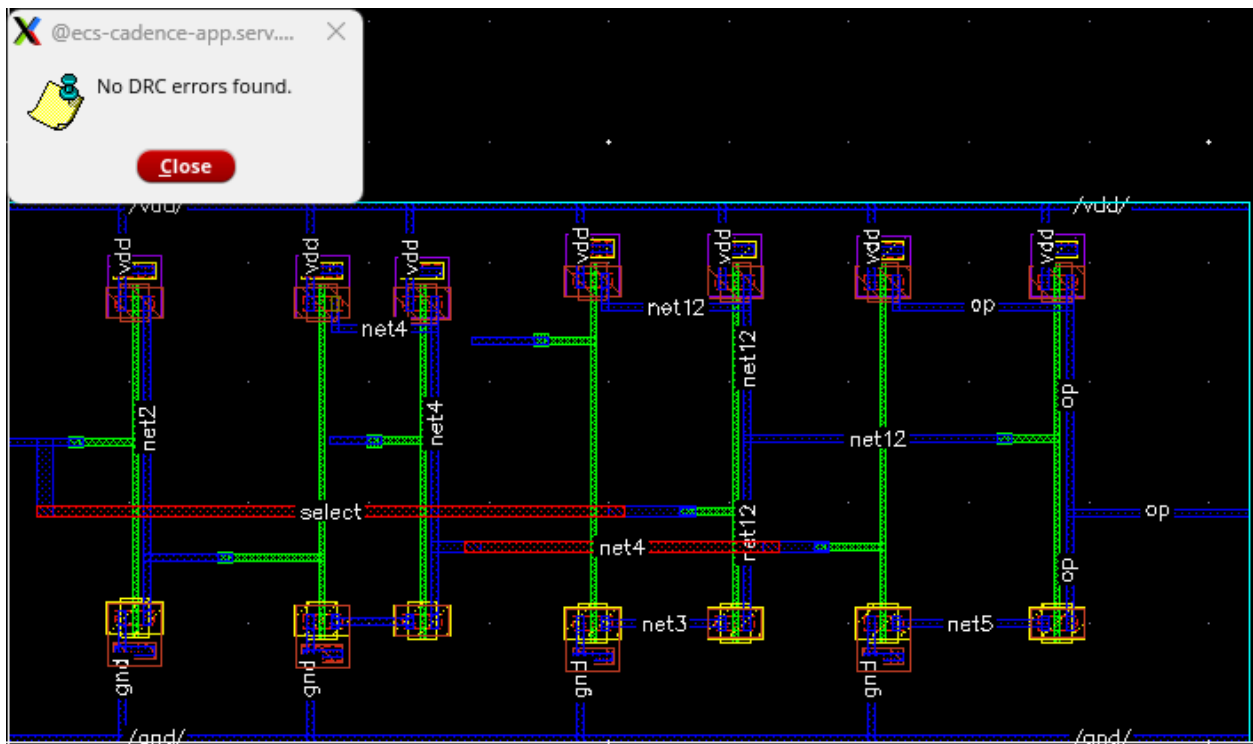
2.0 - Create a Waveform

- Since we use the DRC and LVS to verify that the layout is correct and matches our schematic. We then need to create a waveform. To do this we need to select the Assura Quarus and when that completes a .dspf file will be created with the information for the waveform. We need to go to the maestro tap from the testbench and under setup select the simulation files which we can then upload the .dspf file. This will automatically select inputs and outputs to check and then run the simulation. With this a waveform will be created and then we can check that the waveform is correctly showing the outputs and inputs.

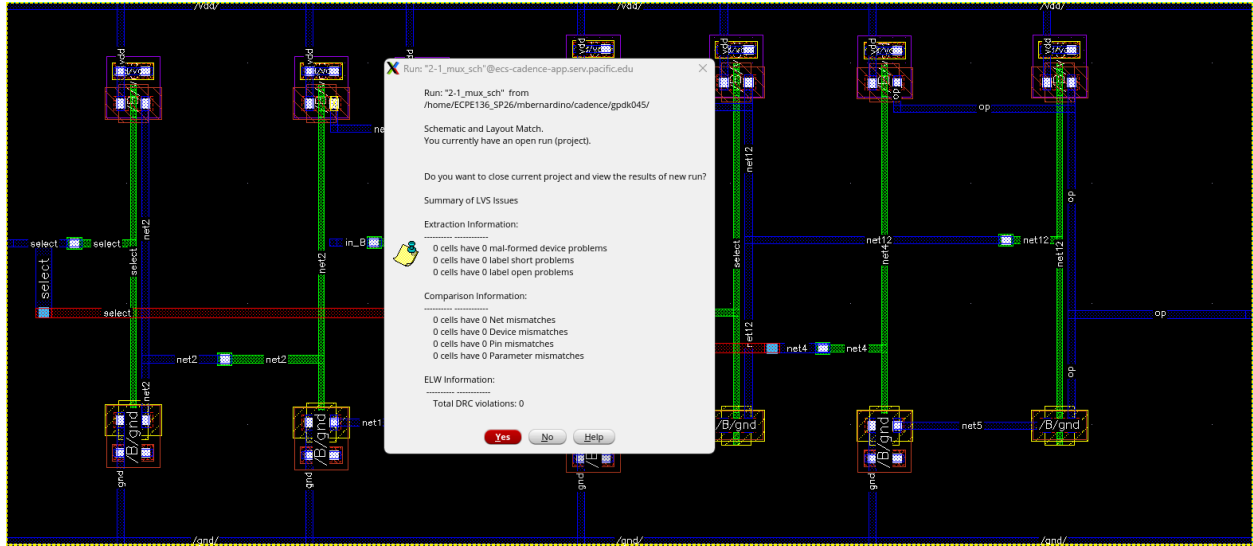
Design description:



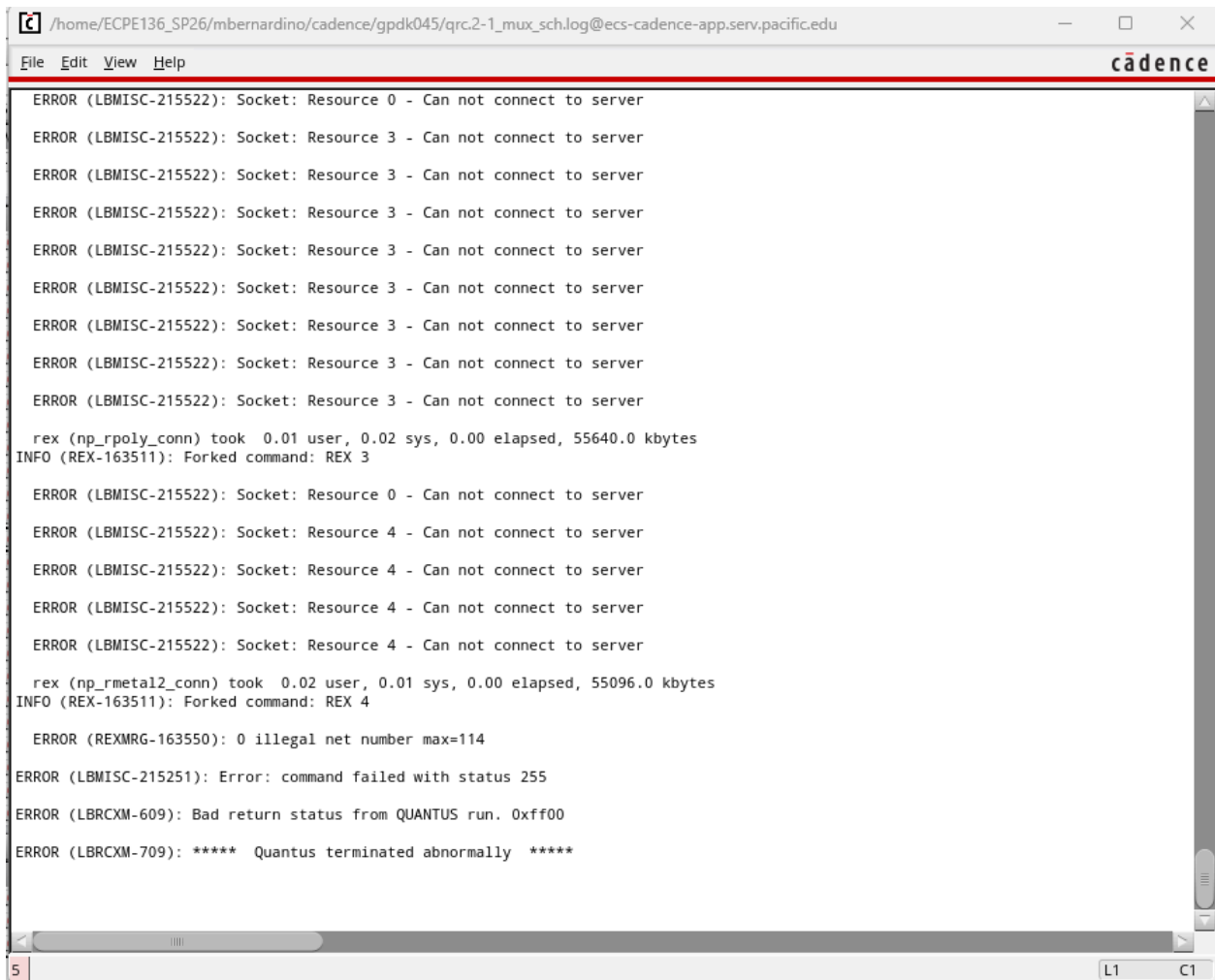
2:1 Mux Layout



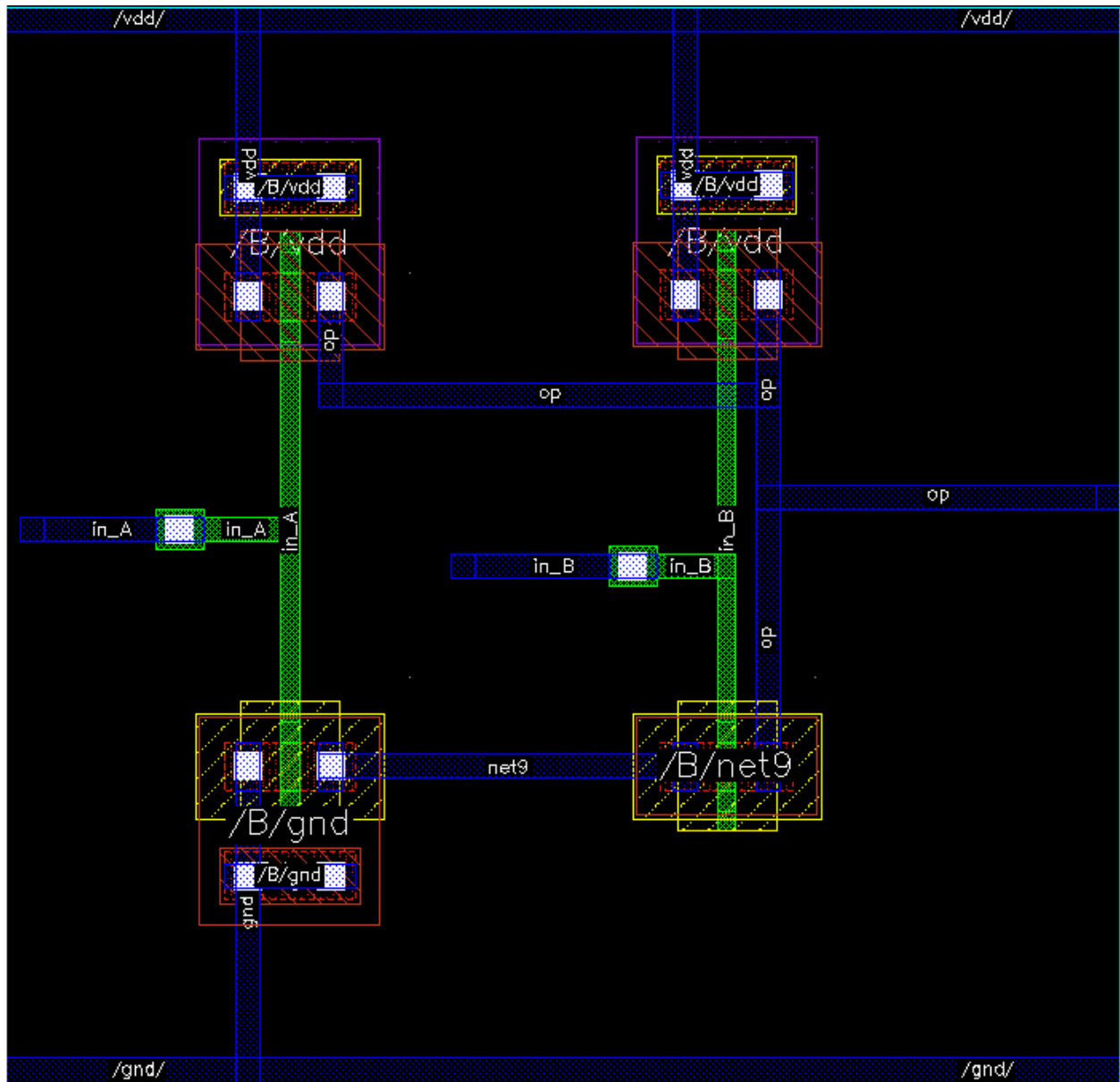
2:1 Mux DRC results (no errors)



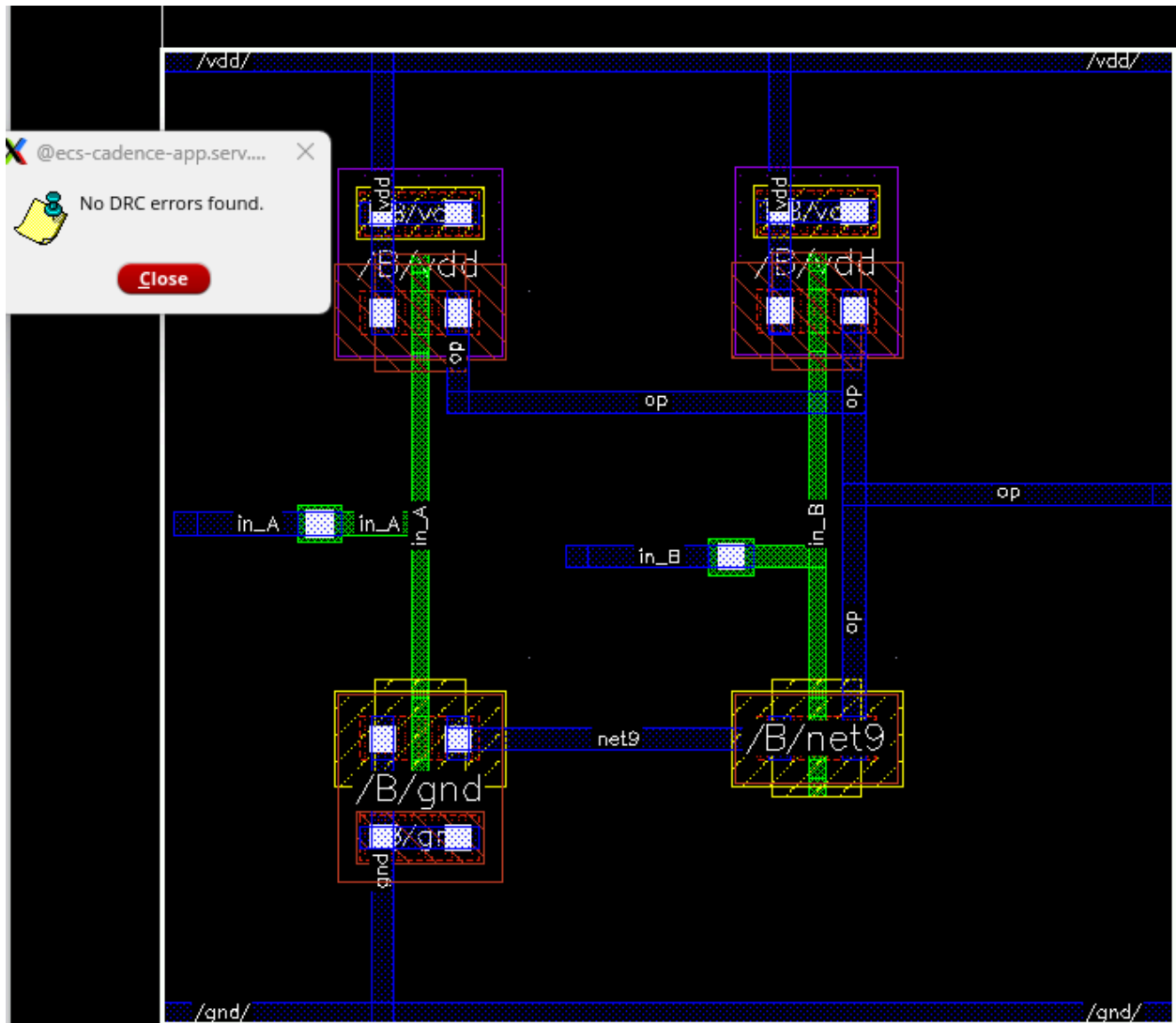
2:1 Mux LVS results (no error)



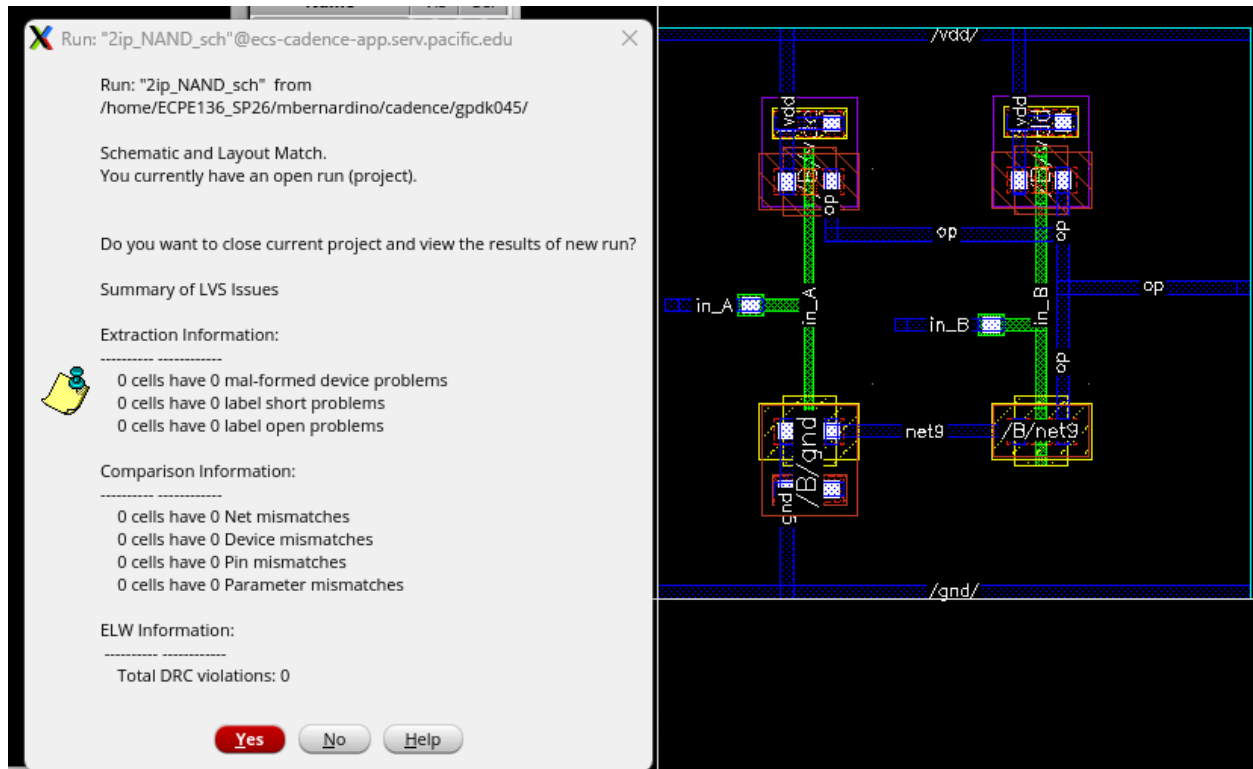
2:1 Mux Quarus Assura (cant connect to server error)



2-Input NAND Layout



2-Input NAND DRC results (no errors)



2-Input NAND LVS results (no error)

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rex (np_rmetal1_conn) took  0.01 user, 0.02 sys, 0.00 elapsed, 58564.0 kbytes
INFO (LBMISC-215513): Socket: connection with sd 14 closed

INFO (REX-163511): Forked command: REX 2

rex (np_rpoly_conn) took  0.01 user, 0.01 sys, 1.00 elapsed, 57256.0 kbytes
INFO (REX-163511): Forked command: REX 3

ERROR (REXMRG-163550): 0 illegal net number max=23

ERROR (LBMISC-215251): Error: command failed with status 255

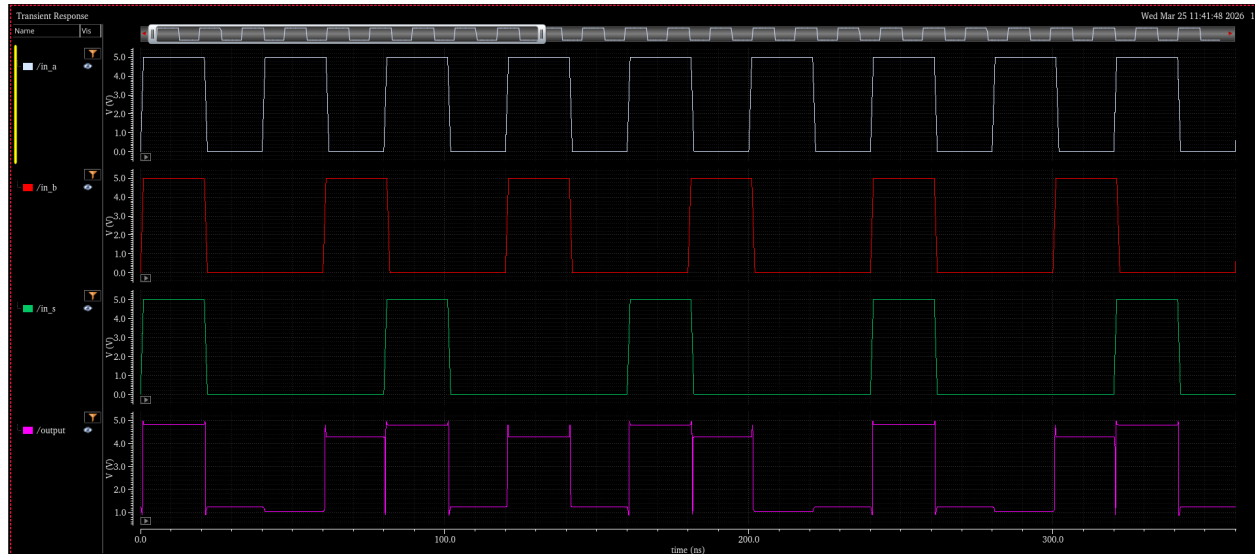
ERROR (LBRCXM-609): Bad return status from QUANTUS run. 0xff00

ERROR (LBRCXM-709): ***** Quantus terminated abnormally *****

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2-Input NAND Quarus Assura (cant connect to server error)

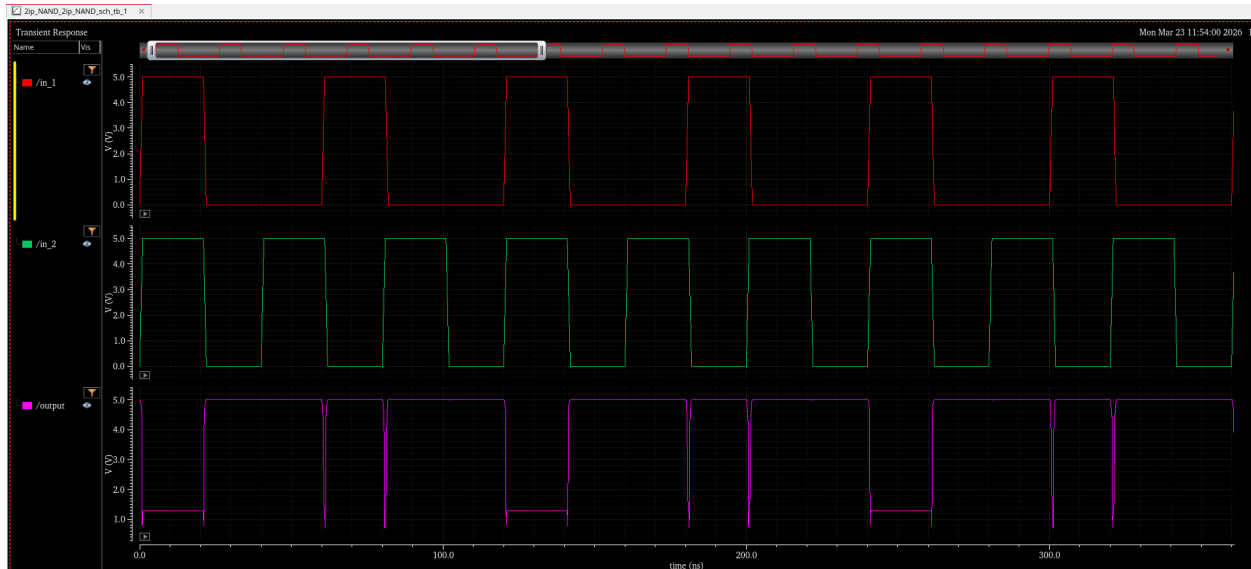
Results & Observations:



2:1 Mux Waveform

Observation

- Since the quarus did not work due to server issues. I am attaching what the waveform should look like from the previous lab assignment. When we look at the waveform we can see three inputs. The main driver for this mux is the select variable. S is the select and when select is low (0) then input B is followed for the output. When select is high (1), then input A is followed for the output.



2-Input NAND Waveform

Observation

- Quarus again still failed due to server issues so I will attach the waveform from the previous assignment as to how it should look. You can see two inputs A and B as well as one Output. When output is low this means both nmos are on (1) and there is a path to ground from the output. When there is only one pmos on (0) then there is a path to vdd from output and we see a high in the graph. Since the pmos are in parallel only one needs to be on to have vdd as output, while; both nmos need to be on since series to have a connection to ground for there to be this output.

Conclusion:

- The goal of this lab was to learn how to merge multiple nand gates together to make a more complex circuit. This complex circuit is a 2:1 mux that takes two inputs with a select input and has one output. A 2:1 mux output reflects whichever input it selects from a and b to then replicate it on the output. We needed to learn and be careful when connecting multiple nand gates as we had a total of three and one inverter. Each a and b input was paired with either select' or select. When making connections we needed to utilize metal 1 and metal 2 so that metals would not interfere with one another especially since we needed to drive the outputs to the next nand gate. This was a learning experience to further progress with cadence virtuoso and layout design. On top of this we created a two input nand gate layout that just simply does that. It was more of a practice before reusing two input nand gates with the more complex 2:1 mux design.